

★ = flagged as cheatsheet material.

Revision

- ★ Exponential distribution: $f(t) = \lambda e^{-\lambda t}$, $t \geq 0$
- ★ Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$
- Expectation of cross-product terms tends to be 0: $E(X_{i,j})$
- ★ To show the random walk does not return to 0 infinitely often (V = count of returns to 0, $E(N)$ = expected number of returns):

$$V = \sum_{n=1}^{\infty} \mathbf{1}\{S_{2n} = 0\}, \quad E[N] = \sum_{n=1}^{\infty} P(S_{2n} = 0) = \sum_{n=1}^{\infty} a_n < \infty$$

- ★ Geometric: $P\{Y = k\} = (1-p)^{k-1}p$, $E[Y] = \frac{1}{p}$
- How to show optional stopping time? If given a condition, flip the logic; apply the flipped logic to $X_1, \dots, X_{T-1}$, but the original logic to X_T .
- ★ $X_1 \perp X_2 \Rightarrow E[X_1 X_2] = E[X_1]E[X_2]$
- ★ **Stopping time:** $\{T = n\}$ is $\mathcal{F}_n$ -measurable — you can tell whether you stopped at time n using only information available at time n , no peeking at the future.
- ★ OST to get the probability that S_n ever reaches X :
 - lower bd $< M_{n \wedge T}$ < upper bd

#	Example	Why it's a martingale	Why it's square integrable	Converges?	$E[M_{\infty}] = E[M_0]?$
1	Coin-flip walk S_n (or $1 + S_n$)	Next step has mean 0	$E[S_n^2] = n$	No	n/a
2	$S_n^2 - n$	$E[S_{n+1}^2 \mathcal{F}_n] = S_n^2 + 1$	$ M_n \leq n^2$	No	n/a
3	Stopped walk $S_{n \wedge T}$, where T = first hit of 1	A stopped martingale is a martingale	$ M_n \leq n$	Yes, to 1	No ($1 \neq 0$)
4	Gambler's ruin $1 + S_{n \wedge T}$, where T = first hit of 0	Same as 3	$0 \leq M_n \leq 1 + n$	Yes, to 0	No ($0 \neq 1$)
5	Doubling strategy	Every bet is fair	$ W_n \leq 2^n$	Yes, to 1	No ($1 \neq 0$)
6	Product martingale $X_1 \cdots X_n$ with $E[X] = 1$ (e.g. $\times 5$ or $\div 5$, your Q3)	$E[M_{n+1} \mathcal{F}_n] = M_n E[X] = M_n$	$E[M_n^2] = (E[X^2])^n$	Yes, to 0	No ($0 \neq 1$)
7	Exponential r^{S_n} for a biased walk (your Q4)	$E[r^{X_n}] = 1$ by the choice of r	$ M_n \leq r^{-n}$	Yes, to 0	No
8	Constant $M_n = c$	Trivial	$c^2 < \infty$	Yes	Yes
9	Two-barrier stopped walk (stop at 0 or 7, as in Q4)	Stopped martingale	Bounded between the barriers	Yes	Yes (that's OST)
10	Doob martingale $M_n = E[Y \mathcal{F}_n]$, with $E[Y^2] < \infty$	Tower property	$E[M_n^2] \leq E[Y^2]$ for all n	Yes	Yes

Lecture 1

- If $m < n$: $E[E(Y | \mathcal{F}_n) | \mathcal{F}_m] = E[Y | \mathcal{F}_m]$
 - Linearity: $E[a + bX | \mathcal{F}_t] = a + bE[X | \mathcal{F}_t]$
 - If Z is $\mathcal{F}_n$ -measurable: $E(ZY | \mathcal{F}_n) = ZE[Y | \mathcal{F}_n]$
 - Standardizing the probability of a random walk:
 - $E(X_1) = \dots = E(X_n) = 0$, $\text{Var}(X_1) = \dots = \text{Var}(X_n) = 1$
 - $S_n = X_1 + \dots + X_n \Rightarrow E(S_n) = 0$ and
- $$\text{Var}(S_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = \underbrace{1 + \dots + 1}_{n \text{ times}} = n$$
- **NOTE:** $|S_n| \sim \sqrt{n}$, since $\sigma = \sqrt{\text{Var}(S_n)} = \sqrt{n}$
 - Hence $\lim_{n \rightarrow \infty} \Pr\left\{a \leq \frac{S_n}{\sqrt{n}} \leq b\right\} = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$
 - **Tower property:** if $m < n$, $E[E(Y | \mathcal{F}_n) | \mathcal{F}_m] = E[Y | \mathcal{F}_m]$
 - If $\mathcal{F}_n$ has no useful information about Y : $E[Y | \mathcal{F}_n] = E[Y]$
 - Probability a random walk returns to the origin: $\mathbb{P}\{S_{2n} = 0\} \sim \frac{1}{\sqrt{\pi n}}$
 - Expected number of steps from the origin until hitting $-N$ or N :
$$e(k) = 1 + \frac{1}{2}e(k-1) + \frac{1}{2}e(k+1), \quad k = -(N-1), \dots, N-1$$
with solution $e(k) = (N+k)(N-k) = N^2 - k^2$
 - ★ Probability of returning to the origin in d dimensions:

$$\mathbb{P}\{S_{2n} = 0\} \sim \frac{c_d}{n^{d/2}}, \quad c_d = \left(\frac{d}{\pi}\right)^{d/2} 2^{1-d}$$

The walker returns to all points finitely often and, if n is large enough, will not return (transient). The walker returns to all points infinitely often when $d = 1$ or 2 (recurrent).

Lecture 2

- 3 types of probability transition matrix: absorbing boundary; reflecting boundary; partially reflecting boundary (sometimes the walker stays at the boundary, other times it leaves).
 - ★ Properties of an irreducible Markov chain:
 - Aperiodic: $\lim_{n \rightarrow \infty} p_n(x, y) = \pi(y)$
 - Periodic: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n p_j(x, y) = \pi(y)$
 - Positive vs. null recurrent: **positive** — solving for the invariant probability (probability vector) gives a solution; **null** — there is no solution.
- ### Lecture 3
- **Martingale:** if $m < n$, $E[M_n | \mathcal{F}_m] = M_m$; equivalently $E[M_{n+1} | \mathcal{F}_n] = M_n$
 - Martingale holds in finite time but fails in infinite time:
 - Discrete stochastic integral example: the bet keeps doubling as long as the player loses via tails; the player wins (game stops) at the first head.
 - Short term $E[M_T] = E[M_0]$. Eventually, with probability 1 the player wins, hence EV is +1.
 - Rational behind optional sampling/stopping:
 - Same example: bet something from the start until T , then bet nothing after T .
 - As long as $T < n$ the martingale holds. If $T > n$ the player keeps losing and we are unsure if he wins, so the martingale may not hold.

Problem Set 1

Conditional Probability

- **Marginal distribution:** e.g. $\Pr(X = i)$, ignoring the other random variables ($Y, Z, \dots$).
- Given a 2D probability matrix (row = X , column = Y), find $E(Y | X = 1)$:

$$\begin{pmatrix} .1 & .1 & 0 & .2 \\ .1 & .05 & .05 & 0 \\ .1 & .1 & .1 & .1 \end{pmatrix}$$

$$((0.1)(1) + (0.1)(2) + (0)(3) + (0.2)(4)) / \Pr(X = 1)$$

- $E[E(X | Y)] = \sum_{i=1}^n E(X | Y = i) \Pr(Y = i)$

– By OST $E[M_T] = E[M_0]$; find this expected value:

$$E[M_T] = (\text{val if } T < \infty)p + (\text{val if } T = \infty)(1-p)$$

– Replace the two values with the bounds.

- ★ **Transient** — may not come back, waiting time unsure. **Null recurrent** — guaranteed to come back, waiting time infinite. **Positive recurrent** — guaranteed to come back, waiting time finite.
- ★ Geometric progression: $\sum_{k=0}^{n-1} r^k = \frac{r^n - 1}{r - 1}$, so $\sum_{k=1}^n 2^k = 2(2^n - 1) = 2^{n+1} - 2$
- ★ Alternative view of $\pi P = \pi$: $\pi(j) = \sum_i \pi(i)p(i, j)$
- ★ Green's function matrix $G(x, y)$ = expected number of visits, starting from x and passing y , before being absorbed per the question's requirements.
- Probability from the Green's function matrix: read the row of G per the question. Identify the correct column by choosing the state that can go to the absorbing state directly. Identify the transition from y in $G(x, y)$ to the destination via $\Pr(y, \text{dest})$ (stated in the question!). Hence choose the specific $G(x, y)$, then multiply $G(x, y) \times \Pr(y, \text{dest})$.
- Conditions for a generator: off-diagonal entries ≥ 0 ; diagonal entries ≤ 0 ; each row sums to 0.
- ★ Counterexamples for martingale assumptions: see table below.

Random Walk (Symmetric and Asymmetric)

- ★ Probability of starting at the origin and returning to the origin — use the binomial theorem formula: $u_{2n} = \binom{2n}{n} p^n q^n$
- ★ Show it does not return to the origin often: impose a limit on the binomial formula to show it converges (reaches a value $< \infty$):

$$\begin{aligned} nC_r &= \frac{n!}{r!(n-r)!} \\ n! &\sim \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n} \\ (2n)! &\sim \sqrt{2\pi (2n)} 2^{2n+\frac{1}{2}} e^{-2n} \\ \sum_{n \geq 1} \frac{1}{n^p} &= \infty \quad \text{iff } p \leq 1 \end{aligned}$$

- A random walk does not return to the origin if the total number of steps is odd.

Symmetric Random Walk

- Standardize to a standard normal: $E(S_n) = 0$, $\text{Var}(S_n) = n$, so s.d. of $S_n = \sqrt{n}$; by CLT $S_n/\sqrt{n} \sim N(0, 1)$.

Gambler's Ruin

- Probability of hitting 0 or N starting from A , without touching A midway:
 - Case 1: start A , end N . Boundaries A, N ; total width $N - A$; probability of a step up to $A + 1$ is $1/(N - A)$.
 - Case 2: start A , end 0. Boundaries 0, A ; total width A ; probability of a step down to $A - 1$ is $1/A$.
 - Both cases occur w.p. 0.5 by symmetry: $0.5 \cdot \frac{1}{N-A} + 0.5 \cdot \frac{1}{A}$
- Expected value of the walk reaching A before hitting 0 or N , starting from A :
 - Same two cases as above, giving $p = 0.5 \cdot \frac{1}{N-A} + 0.5 \cdot \frac{1}{A}$
 - By the geometric distribution, it asks “how many coin flips (or passes through A) before escaping to either boundary?” Hence $1/p$.
- Expected value of the walk reaching B before hitting 0 or N , starting from A :
 - Part 1: probability of starting at A and ending at B . Visualize boundaries 0 and B , ignore N for now: A/B .
 - Part 2: probability of starting at B and ending at 0 or N . Case 1: boundaries B, N , width $N - B$, step up $1/(N - B)$. Case 2: boundaries 0, B , width B , step down $1/B$. By symmetry $0.5 \cdot \frac{1}{N-B} + 0.5 \cdot \frac{1}{B}$.
 - Overall: $\frac{A}{B} \left(0.5 \cdot \frac{1}{N-B} + 0.5 \cdot \frac{1}{B}\right)$

Problem Set 2

Markov Chain

- Aperiodic / irreducible, and proving both at once:
 - **Aperiodic** = (for an irreducible chain) some diagonal entry is non-zero, i.e. a self-loop $p(i, i) > 0$ exists, so the period is 1.
 - **Irreducible** = for all i, j it is possible to go from i to j and from j to i .
 - All entries in the transition probability matrix are non-zero.
- **Invariant probability / probability vector:** if the transition matrix is aperiodic and irreducible then $\bar{\pi}P = \bar{\pi}$ applies; each entry of $\bar{\pi}$ is the fraction of time spent in that state.
- **Bayes:** $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$ — useful when the conditional probability is such that condition B happens at a later time.
- Aggregate probability of 3 transitions from A to C : multiply the transition matrix by itself 3 times, then read the row/column for A and C .
- Expected time from state S back to state S : from the invariant probability, find the probability for state S and take the inverse.
- Expected number of visits to B , starting from A , ending at C : Green's function. Rearrange rows/columns so B is the very first entry. Make B absorbing by setting its probability to 1 in that row only (that column is unaffected). Identify Q — the submatrix with no rows or columns touching B . Solve $G = (I - Q)^{-1}$. G_{ij} is the expected number of visits to j (B) starting from i (A) before ending at C .

- Expected number of visits to B or B' , starting from A , ending at C : as above but put B and B' in the first and second diagonal entries and make each absorbing. Q excludes all rows/columns touching B or B' . Solve $G = (I - Q)^{-1}$, sum all columns in each row, and read the row for starting state A .
- Expected time from state A to state B :
 - Method 1:** green book / interview prep method — solve the system of linear equations.
 - Method 2:** Green's function — put B first, make it absorbing, form Q , solve $G = (I - Q)^{-1}$, sum all columns in each row, read the row for A .
- Doubly stochastic:** every row and every column of the transition matrix sums to 1 (then the uniform π is invariant).

Problem Set 3

Polya's Urn???

Martingale

- ★ What is a martingale / how to prove a random walk is one:
 - Sequence $X_1 \dots X_n$ associated with $\mathcal{F}_n$; both a sub- and super-martingale.
 - $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$
 - There is $\mathcal{F}$ -measurability.
 - for $\forall j$, $\mathbb{E}[M_j] < \infty$
- Sub-martingale:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$. **Super-martingale:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \leq M_n$.
- Discrete stochastic integral and required steps:
 - The summation of the martingale can be re-expressed as a summation of bets and the change in the random variable. A player cannot win a fair game with a betting strategy in a finite amount of time: $\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] = W_n$.
 - B_n is $\mathcal{F}_{n-1}$ -measurable.
 - Prove the bet is bounded by a constant: $|B_n| \leq C_n$.
- Proving stopping time:
 - $\{T = n\} = \{X_1 = \cdot, X_2 = \cdot, \dots, X_n = \cdot\}$
 - X_1 to X_{n-1} do not fulfill the condition, while X_n fulfills it and the game stops.
 - It is not a stopping time when a random variable is outside of the information set.
- Optional stopping theory:
 - $M_{n \wedge T}$ is a martingale for any stopping time T .
 - Universal truth, applies no matter what: $\mathbb{E}[M_{n \wedge T}] = \mathbb{E}[M_0]$ for every fixed n .
 - If T is bounded, use $M_{k \wedge T} = M_T$.
 - Is the stopped process bounded? Verify $\mathbb{E}[|M_{n \wedge T}|^2] \leq C$, and check $\mathbb{P}(T < \infty) = 1$ and $\mathbb{E}[M_T] < \infty$.
 - Not the above 2 bullets? Verify $\lim_n \mathbb{E}[|M_n| \mathbf{1}\{T > n\}] = 0$, and check $\mathbb{P}(T < \infty) = 1$ and $\mathbb{E}[M_T] < \infty$.
 - Therefore $\mathbb{E}[M_T] = \mathbb{E}[M_0]$ — most important!

Problem Set 4

- ★ **Martingale convergence theorem:** suppose M_n is a martingale w.r.t. $\{\mathcal{F}_n\}$ and there exists $C < \infty$ such that $\mathbb{E}[|M_n|] \leq C$ for all n . Then there exists a random variable M_∞ such that with probability one $\lim_{n \rightarrow \infty} M_n = M_\infty$.
Warning: the theorem does *not* give $\mathbb{E}[M_\infty] = \mathbb{E}[M_0]$.
- ★ Martingale betting strategy:
 - $P\{X_j = 1\} = P\{X_j = -1\} = \frac{1}{2}$; let $M_0 = 0$, $M_n = X_1 + \dots + X_n$

- $-W_n = \sum_{j=1}^n B_j \Delta M_j = \sum_{j=1}^n B_j X_j$, $W_0 = 0$
- $-B_1 = 1$ and for $j > 1$: $B_j = 2^{j-1}$ if $X_1 = \dots = X_{j-1} = -1$, $B_j = 0$ otherwise.
- $-B_n$ is $\mathcal{F}_{n-1}$ -measurable (the bet is placed before the flip is seen) and bounded by some $K_n < \infty$, so W_n is a discrete stochastic integral and hence a martingale. In particular $\mathbb{E}[W_n] = 0$ for every n .
- $-W_n = 1$ unless $X_1 = \dots = X_n = -1$; if all losses,

$$W_n = -1 - 2 - 4 - \dots - 2^{n-1} = -[2^n - 1] = 1 - 2^n$$

- with $P\{\text{all losses}\} = 2^{-n}$.
- $-E[W_n] = 1 \cdot [1 - 2^{-n}] - [2^n - 1] \cdot 2^{-n} = 0$

- ★ Poisson: $P\{X = k\} = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$, $E[N] = \text{Var}(N) = \lambda t$
- ★ Poisson mean waiting time for k intervals: $k E[T_j] = \frac{k}{\lambda}$
- ★ “Poisson binomial” — probability that x have arrived at $t = t_1$ and $x + k$ have arrived at $t = t_2$:

$$\binom{x+k}{x} \left(\frac{t_1}{t_1+t_2}\right)^x \left(\frac{t_2}{t_1+t_2}\right)^k$$

- DISCLAIMER:** only works for $\Pr(t = x \mid t = x + k)$, not $\Pr(t = x + k \mid t = x)$.
- To solve $\Pr(t = x + k(A) \mid t = x(B))$, it is equivalent to solving $\Pr(t = k(A))$ because of the disjoint nature.
- ★ Interpreting $P\{X_1(X_6 - X_2) = 0\}$: see it as 2 disjoint intervals $(0, 1)$ and $(2, 6)$; we want either to be 0, hence a union:

$$P\{X_1 = 0\} + P\{X_6 - X_2 = 0\} - P\{X_1 = 0\}P\{X_6 - X_2 = 0\}$$

- ★ “At most k people arrived in x hours”: $P\{X_x \leq k\} = \sum_{j=0}^k P\{X_x = j\}$, then use the Poisson formula for each term.
- Generator matrix from a continuous-time Markov chain, and mean time per state:
 - List all the stated probabilities provided by the question.
 - For diagonal entries, set them so the entry plus the sum of the corresponding row entries equals 0.
 - Mean time in each state i : $1/|\alpha(i, i)|$ (the diagonal entry is negative).
- Continuous-time MC, expected time from state i back to state i (different from discrete time!): find the invariant probability ($\pi A = 0$), then compute $1/(\pi(i) |\alpha(i, i)|)$.
- Square integrable martingale:** M_n is square integrable if $\mathbb{E}[M_n^2] < \infty$ for each n .

$$\mathbb{E}[M_n^2] = \mathbb{E}[M_0^2] + \sum_{j=1}^n \mathbb{E}[(M_j - M_{j-1})^2]$$

$$\begin{aligned} M_n^2 &= \left[M_0 + \sum_{j=1}^n (M_j - M_{j-1})\right]^2 \\ &= M_0^2 + \sum_{j=1}^n (M_j - M_{j-1})^2 \\ &\quad + \sum_{j \neq k} (M_j - M_{j-1})(M_k - M_{k-1}) \end{aligned}$$